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Exploring Non-Negative Matrix Factorization: Algorithms and Applications

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Outline

Aim of the thesis: explore the problem of non-negative matrix factorization (NMF) by analyzing the existing literature on the topic, and then studying and implementing existing algorithms to solve it.

1. Motivation
   - NMF Definition
   - Applications
2. Algorithms
   - MU
   - PGD
   - ALS
   - HALS
3. Implementation and Results
   - Python implementation
   - Numerical simulations

Make Titles Informative. Use Uppercase Letters.

Definition: (NMF problem)

Given a non-negative matrix $V \in \mathbb{R}_+^{m \times n}$ and a factorization rank $r \in \mathbb{N}$, find two non-negative matrices $W \in \mathbb{R}_+^{m \times r}$ and $H \in \mathbb{R}_+^{r \times n}$ that minimize a given cost function $D(V, WH)$, that is:

$$\min_{W \in \mathbb{R}^{m \times r}, H \in \mathbb{R}^{r \times n}} D(V, WH) \tag{1}$$

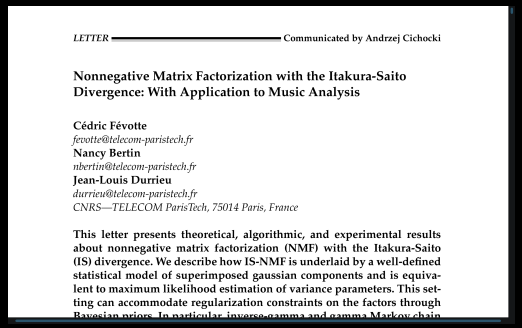
subject to:

$$W \geq 0, H \geq 0$$

NMF is extremely useful for inherently non-negative datasets. For instance, when the data represent physical quantities like signal strengths or pixel intensities.

Example applications include:

- Audio signal processing.
- Feature extraction in images.
- Topic modeling in text mining.



LETTER Communicated by Andrzej Cichocki

Nonnegative Matrix Factorization with the Itakura-Saito Divergence: With Application to Music Analysis

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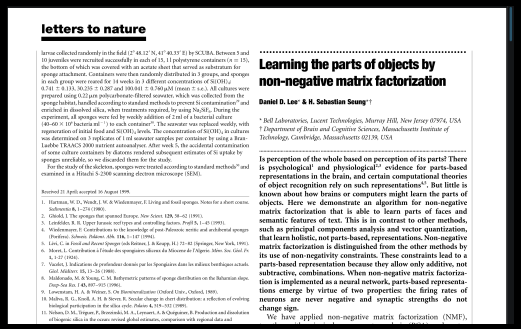
This letter presents theoretical, algorithmic, and experimental results about nonnegative matrix factorization (NMF) with the Itakura-Saito (IS) divergence. We describe how IS-NMF is underlaid by a well-defined statistical model of superimposed gaussian components and is equivalent to maximum likelihood estimation of variance parameters. This setting can accommodate regularization constraints on the factors through Bayesian priors. In particular, inverse-gamma and gamma Markov chain



letters to nature

larvae collected randomly in the field (2° 48.12′ N, 41° 40.33′ E) by SCUBA. Between 5 and 10 juveniles were recruited successfully in each of 15, 11 polystyrene containers ($n \geq 15$), the bottom of which was covered with an acetate sheet that served as substratum for sponge attachment. Containers were then randomly distributed in 3 groups, and sponges in each group were reared for 14 weeks in 3 different concentrations of $Si(OH)_4$: 0.741 ± 0.133, 30.259 ± 0.267 and 100.041 ± 0.760 μM (mean ± s.e.). All cultures were prepared using 0.22 μm polycarbonate-filtered seawater, which was collected from the sponge habitat. Installed according to standard methods to prevent Si contamination[7] and enriched in dissolved silica, when treatments required, by using Na_2SiF_6. During the experiment, all sponges were fed by weekly addition of 2 ml of a bacterial culture ($40–60 \times 10^6$ bacteria ml^{-1}) to each container[8]. The seawater was replaced weekly, with regeneration of initial food and $Si(OH)_4$ levels. The concentration of $Si(OH)_4$ in cultures was determined on 5 replicates of 1 ml seawater samples per container by using a Bran-Luebbe TRAACS 2000 nutrient autoanalyser. After week 5, the accidental contamination of some culture containers by diatoms rendered subsequent estimates of Si uptake by sponges unreliable, so we discarded them for the study.

For the study of the skeleton, sponges were treated according to standard methods[9] and examined in a Hitachi S-2500 scanning electron microscope (SEM).

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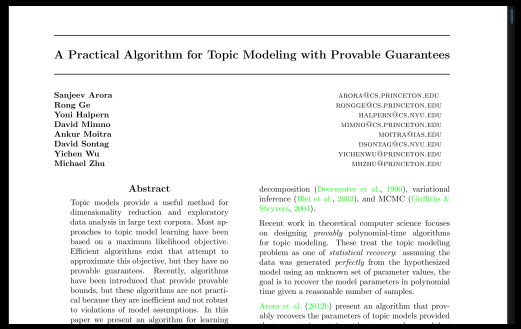

Learning the parts of objects by non-negative matrix factorization

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Is perception of the whole based on perception of its parts? There is psychological[1] and physiological[2,3] evidence for parts-based representations in the brain, and certain computational theories of object recognition rely on such representations[4,5]. But little is known about how brains or computers might learn the parts of objects. Here we demonstrate an algorithm for non-negative matrix factorization that is able to learn parts of faces and semantic features of text. This is in contrast to other methods, such as principal components analysis and vector quantization, that learn holistic, not parts-based, representations. Non-negative matrix factorization is distinguished from the other methods by its use of non-negativity constraints. These constraints lead to a parts-based representation because they allow only additive, not subtractive, combinations. When non-negative matrix factorization is implemented as a neural network, parts-based representations emerge by virtue of two properties: the firing rates of neurons are never negative and synaptic strengths do not change sign.

We have applied non-negative matrix factorization (NMF),

A Practical Algorithm for Topic Modeling with Provable Guarantees

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Abstract

Topic models provide a useful method for dimensionality reduction and exploratory data analysis in large text corpora. Most approaches to topic model learning have been based on a maximum likelihood objective. Efficient algorithms exist that attempt to approximate this objective, but they have no provable guarantees. Recently, algorithms have been introduced that provide provable bounds, but these algorithms are not practical because they are inefficient and not robust to violations of model assumptions. In this paper we present an algorithm for learning

decomposition (Deerwester et al., 1990), variational inference (Blei et al., 2003), and MCMC (Griffiths & Steyvers, 2004).

Recent work in theoretical computer science focuses on designing *provably* polynomial-time algorithms for topic modeling. These treat the topic modeling problem as one of *statistical recovery*, assuming the data was generated *perfectly* from the hypothesized model using an unknown set of parameter values, the goal is to recover the model parameters in polynomial time given a reasonable number of samples.

Arora et al. (2012b) present an algorithm that provably recovers the parameters of topic models provided

Example: learning parts of objects with NMF (Lee & Seung [2])

Figure: Sample images from the CBCL face dataset.

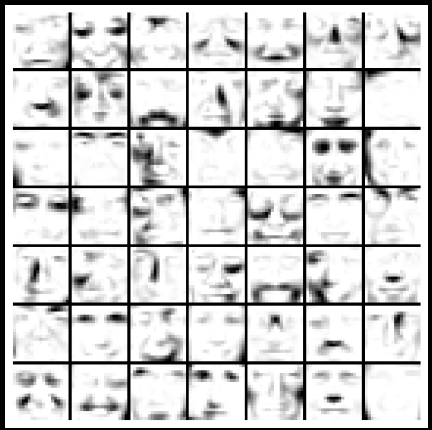

Figure: Basis images obtained from NMF.

- CBCL face dataset $\rightarrow$ 2429 images of faces with 19x19 pixels.
- NMF with $r = 49 \rightarrow 49$ basis images and corresponding coefficients.
- Basis images extract facial features.
- Columns of H contain the weight of each feature for each image.
- Original images reconstructed as $V[:, i] \approx \sum_{k=1}^{r} W[:, k]H[k, i]$.

Lee and Seung proposed the Multiplicative Updates (MU).

Can be interpreted as a scaled gradient descent method:
the gradient of the cost function with respect to H is given by:

$$\nabla_H \frac{1}{2}\|V - WH\|_F^2 = W^T WH - W^T V \tag{2}$$

taking a step of size:

$$\eta_k = \frac{H}{W^T WH} \tag{3}$$

yields the multiplicative update rule:

MU for H

$$H \leftarrow H \circ \frac{W^T V}{W^T WH} \tag{4}$$

Multiplicative Updates (MU) for NMF

Algorithm 1 Multiplicative Update for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**

$H^{(k+1)} \leftarrow H^{(k)} \circ \frac{W^{(k)\top} V}{W^{(k)\top} W^{(k)} H^{(k)}}$ ▷ Update for H

$W^{(k+1)} \leftarrow W^{(k)} \circ \frac{V H^{(k+1)\top}}{W^{(k)} H^{(k+1)} H^{(k+1)\top}}$ ▷ Update for W

end while

return W, H

Projected Gradient Descent (PGD) for NMF

Algorithm 2 Projected Gradient Descent for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**

  $\mathrm{LW} \leftarrow \|W^{(k)}\|_2^2$

  $\mathrm{LH} \leftarrow \|H^{(k)}\|_2^2$ ▷ Lipschitz constants

  $\mathrm{Grad_W} \leftarrow W^{(k)} H^{(k)} H^{(k)\top} - V H^{(k)\top}$ ▷ Gradient w.r.t. W

  $W^{(k+1)} \leftarrow \max\left(0, W^{(k)} - \frac{1}{LW}\mathrm{Grad_W}\right)$ ▷ Update for W

  $\mathrm{Grad_H} \leftarrow W^{(k+1)\top} W^{(k+1)} H^{(k)} - W^{(k+1)\top} V$ ▷ Gradient w.r.t. H

  $H^{(k+1)} \leftarrow \max\left(0, H^{(k)} - \frac{1}{LH}\mathrm{Grad_H}\right)$ ▷ Update for H

end while

return W, H

Alternating Least Squares (ALS) for NMF

Algorithm 3 Alternating Least Squares for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**

  for i = 1 to m **do** ▷ m independent linear least squares

    $W(i,:) \leftarrow \max\left(0, \arg\min_w \|V(i,:) - wH^{(k)}\|_2^2\right)$

  end for

  for j = 1 to n **do** ▷ n independent linear least squares

    $H(:,j) \leftarrow \max\left(0, \arg\min_h \|V(:,j) - W^{(k)}h\|_2^2\right)$

  end for

end while

return W, H

Hierarchical Alternating Least Squares (HALS) for NMF

Algorithm 4 Hierarchical Alternating Least Squares for NMF

Require: $V \in \mathbb{R}_+^{m \times n}$, rank r, initial factors $W^{(0)} \in \mathbb{R}_+^{m \times r}$, $H^{(0)} \in \mathbb{R}_+^{r \times n}$

while stopping criterion not met **do**
  for $\ell = 1$ to r **do**
    $\tilde{R}_\ell \leftarrow V - \sum_{k \neq \ell} W[:, k] H[k, :]$ ▷ Residual matrix w.r.t. ℓ
    $H[\ell, :] \leftarrow \max\left(0, \frac{\tilde{R}_\ell^\top W[:,\ell]}{W[:,\ell]^\top W[:,\ell]}\right)$
  end for
  for $\ell = 1$ to r **do**
    $\tilde{R}_\ell \leftarrow V - \sum_{k \neq \ell} W[:, k] H[k, :]$
    $W[:, \ell] \leftarrow \max\left(0, \frac{\tilde{R}_\ell H[\ell,:]^\top}{H[\ell,:] H[\ell,:]^\top}\right)$
  end for
end while
return W, H

The Algorithms have been implemenetd in Python

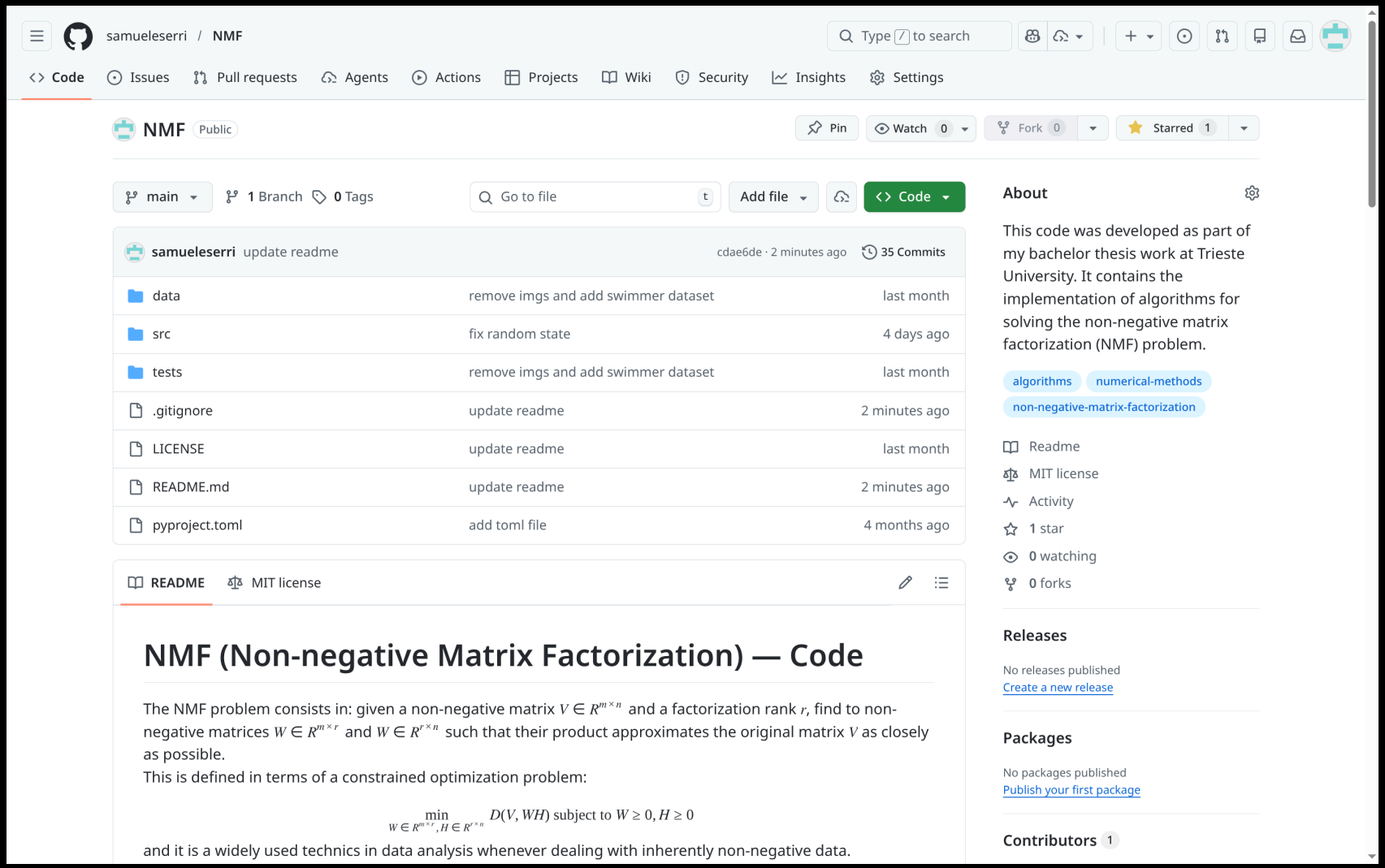

Figure: https://github.com/samueleserri/NMF.git

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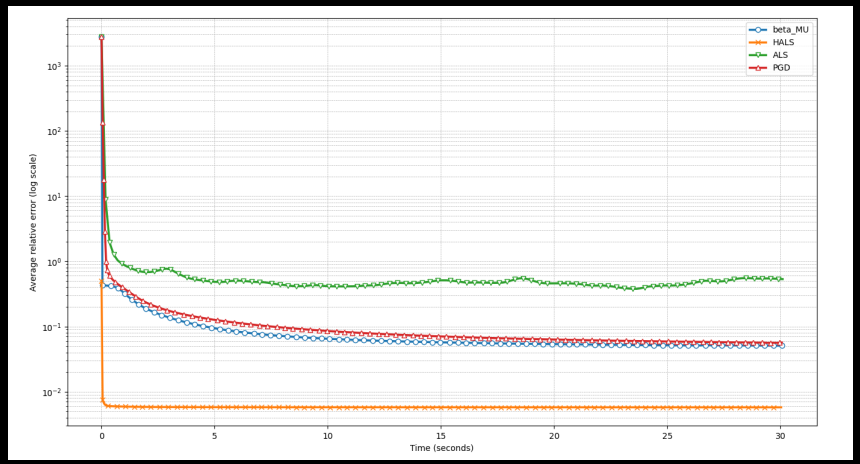

Figure: Average relative reconstruction error for each algorithm averaged over 10 random initializations on the CBCL dataset.

Numerical simulations on the CBCL face dataset with rank $r = 49$, averaged over 10 random initialization. Each algorithm was run for 30 seconds, and the average number of iterations and time per iteration were recorded.

Algorithm	Avg. iterations	Avg. time/iter (s)
MU	324	9.26×10^{-2}
HALS	362	8.29×10^{-2}
ALS	144	2.08×10^{-1}
PGD	282	1.06×10^{-1}

Table: Average iterations and time per iteration for each algorithm.

Results

HALS converges the fastest and reaches the lowest relative reconstruction error within the 30 seconds. ALS perform the worst, with the highest time per iteration and lowest number of iterations. PGD has a higher time per iteration than MU and HALS, but converges faster than ALS. MU and PGD have similar performance.

For Further Reading and References I

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Thank you for your attention!